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(54) **MEDICAL ELONGATED BODY AND
MEDICAL INSTRUMENT SET**

(71) Applicant: **TERUMO KABUSHIKI KAISHA,**
Tokyo (JP)

(72) Inventors: **Kenta Suzuki,** Fujinomiya (JP); **Kota
Hamuro,** Fujinomiya (JP); **Takito
Inukai,** Fujinomiya (JP)

(73) Assignee: **TERUMO KABUSHIKI KAISHA,**
Tokyo (JP)

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CPC **A61B 17/3423** (2013.01); **A61B
2017/00292** (2013.01); **A61B 2017/00778**
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(58) **Field of Classification Search**

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International Application No. PCT/JP2018/012430.

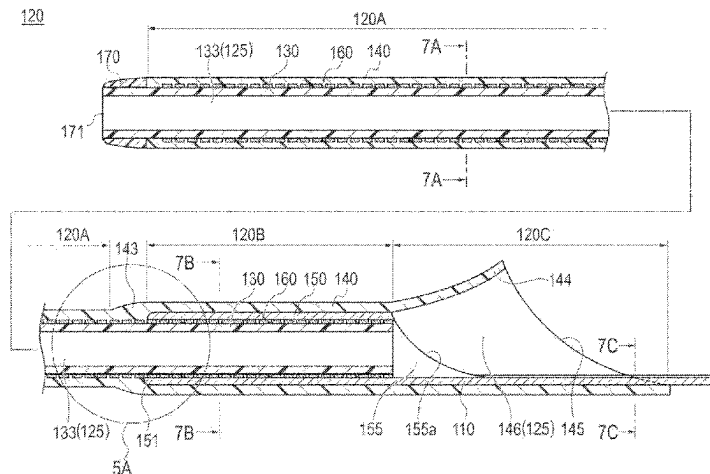
Primary Examiner — Richard G Louis

(74) *Attorney, Agent, or Firm* — Buchanan Ingersoll &
Rooney PC

(57) **ABSTRACT**

A medical elongated body including a shaft portion com-
prising a linear body and a proximal portion; a distal
member disposed in a distal side of the shaft portion and
having a distal end and a proximal end, the distal member
having a first region having an inner layer and an outer layer
which are made of a resin material, and a second region
having the inner layer and the outer layer, and a tubular
member made of a metallic material and disposed between
the inner layer and the outer layer in the second region, the
proximal end opening portion comprising an inclined region
having a proximally-decreasing dimension; and a reinforce-
ment layer comprising a metallic material arranged between

(Continued)



the inner layer and the outer layer from the first region to the second region of the distal member.

20 Claims, 7 Drawing Sheets

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continuation of application No. 16/583,411, filed on Sep. 26, 2019, now Pat. No. 11,134,982, which is a continuation of application No. PCT/JP2018/012430, filed on Mar. 27, 2018.

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(52) U.S. Cl.

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(58) Field of Classification Search

CPC *A61M 25/0068*; *A61M 25/0108*; *A61M 25/005*; *A61M 31/005*; *A61M 25/0097*; *A61M 25/0662*; *A61M 2025/0059*; *A61M 2025/0183*; *A61M 2025/0681*; *A61M 25/01*

See application file for complete search history.

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FIG. 1

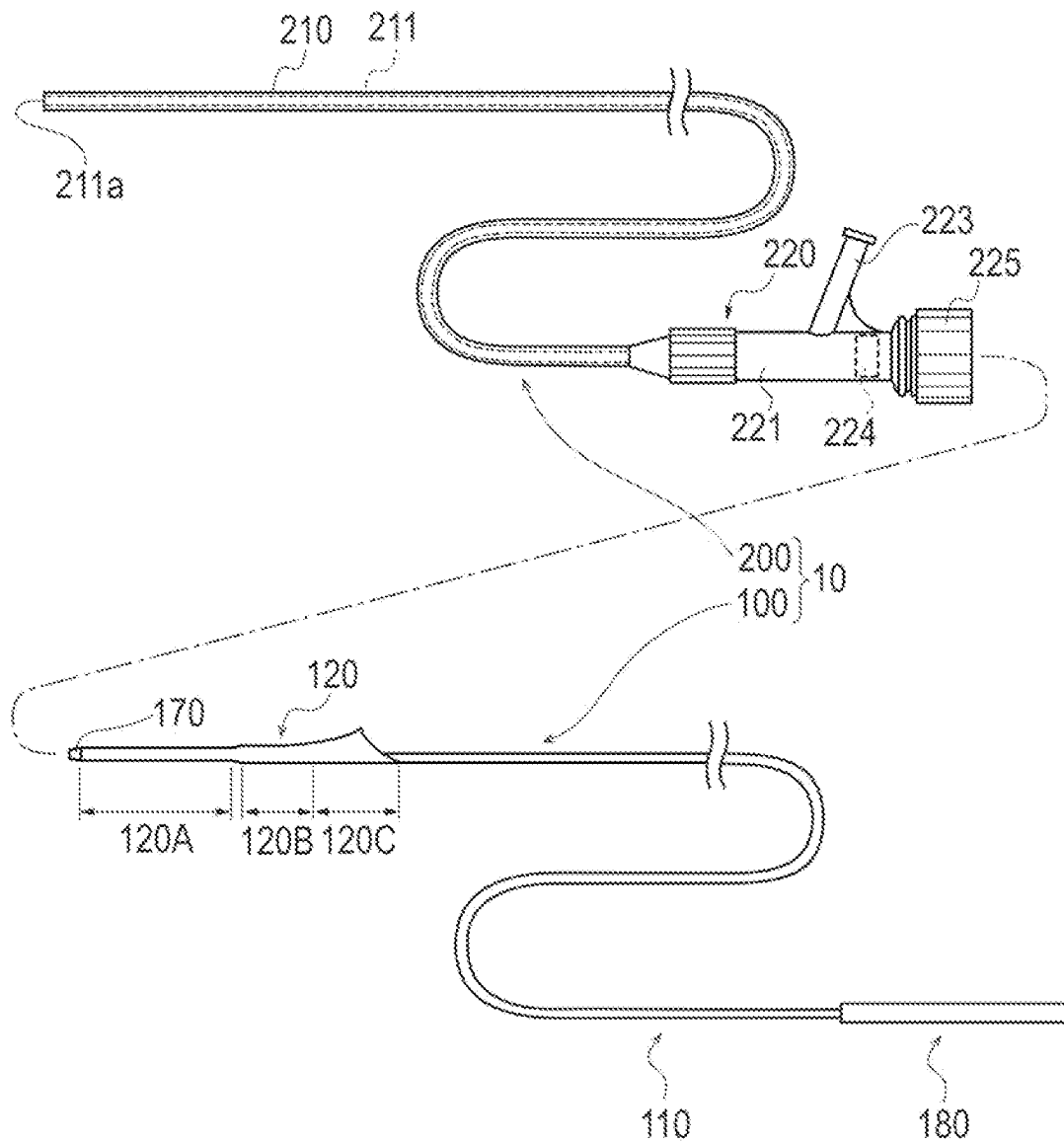


FIG. 2

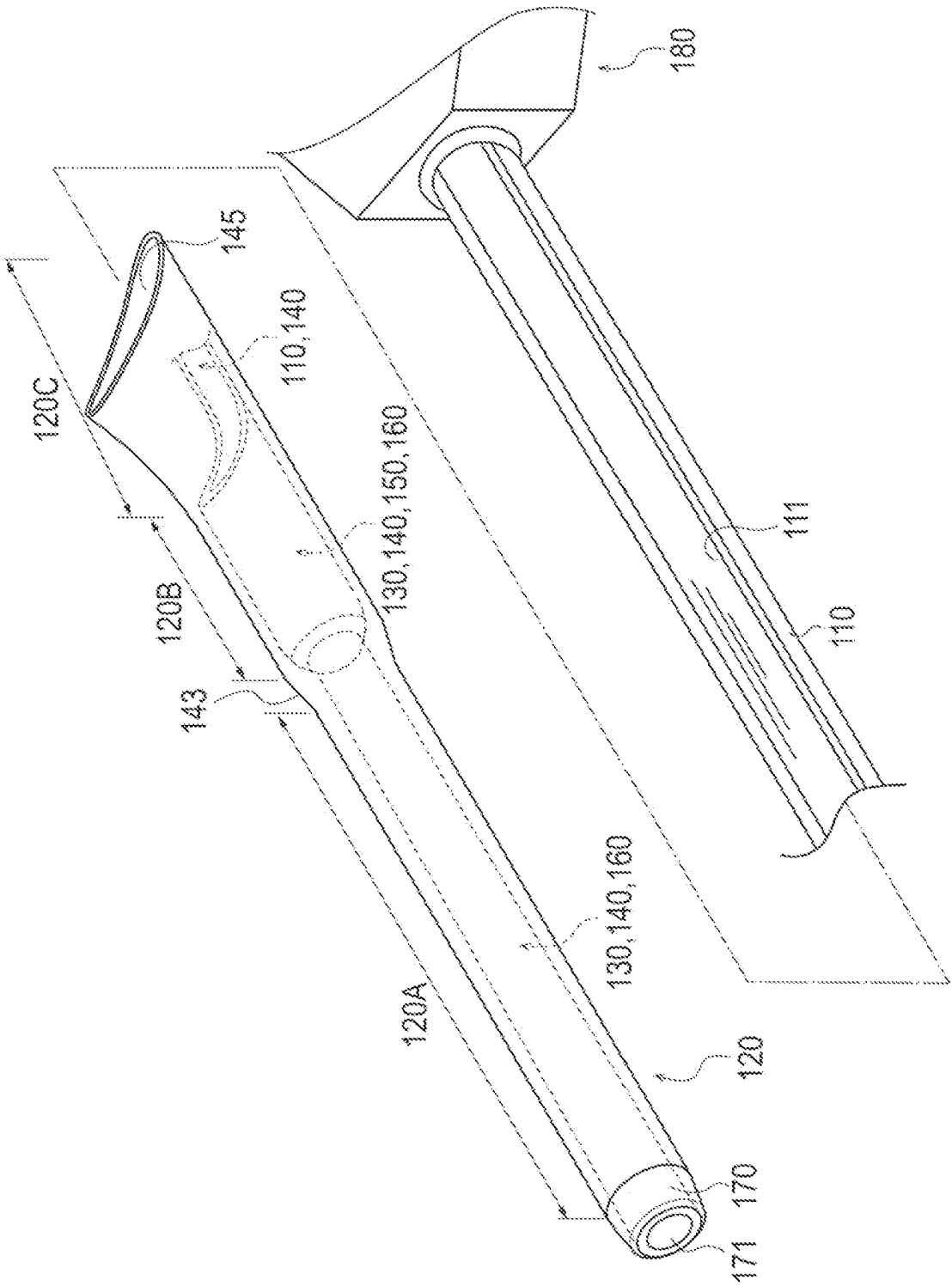


FIG. 3

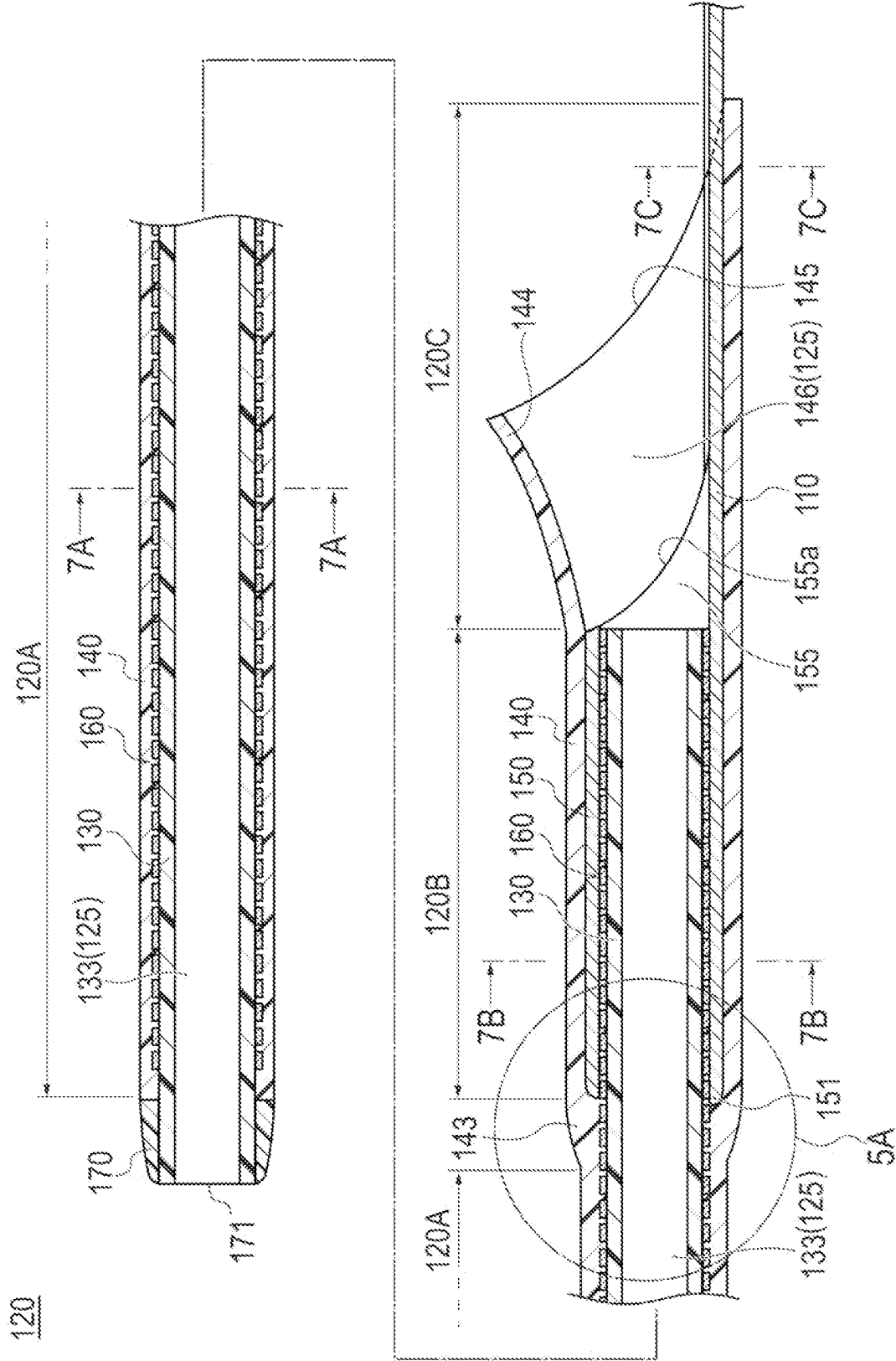


FIG. 4

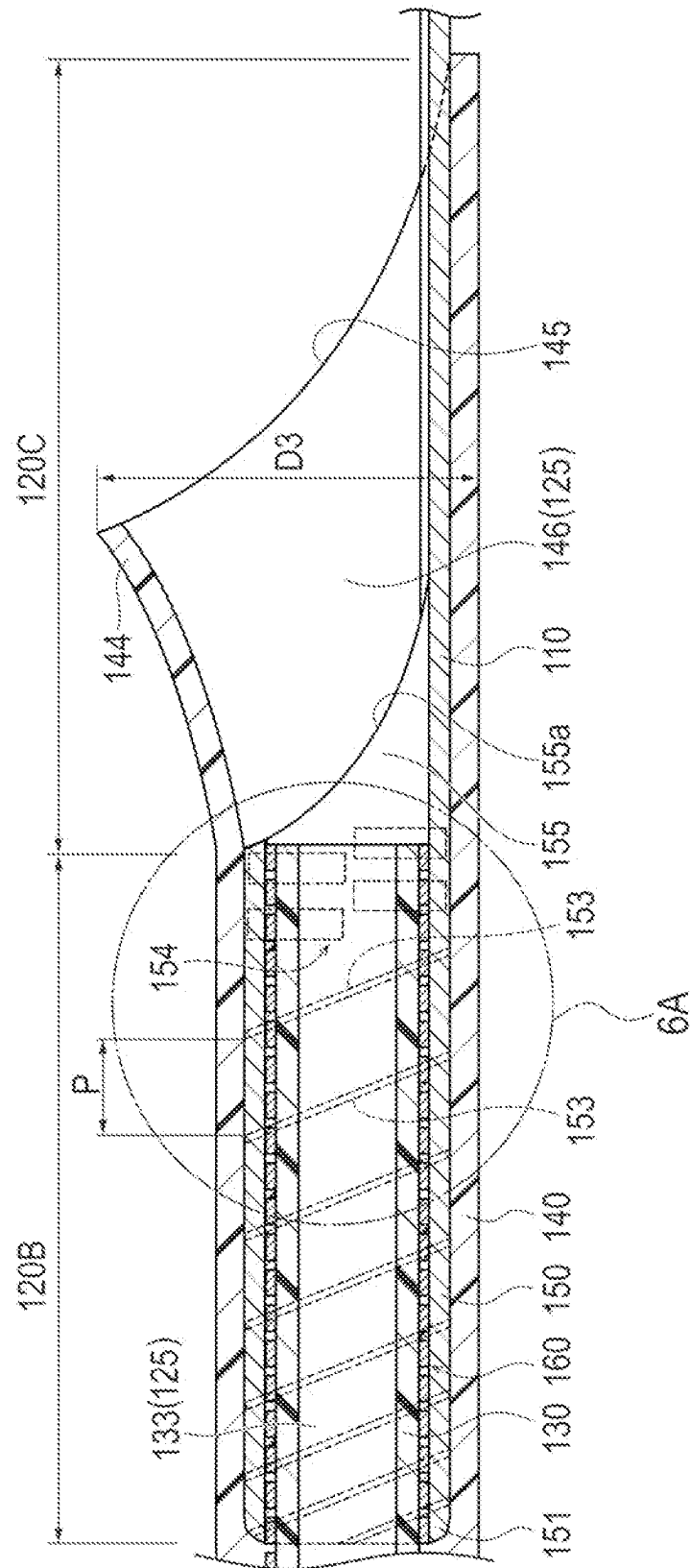


FIG. 5

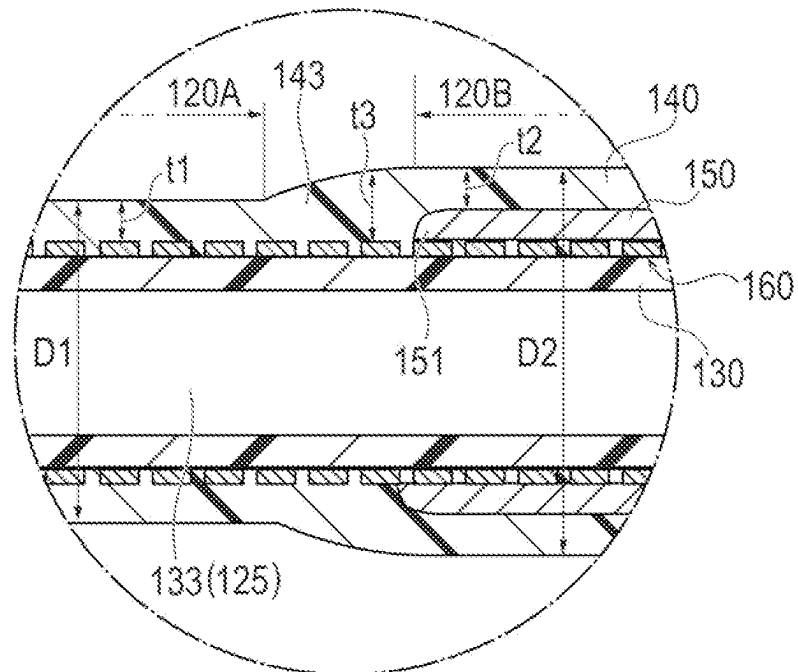


FIG. 6

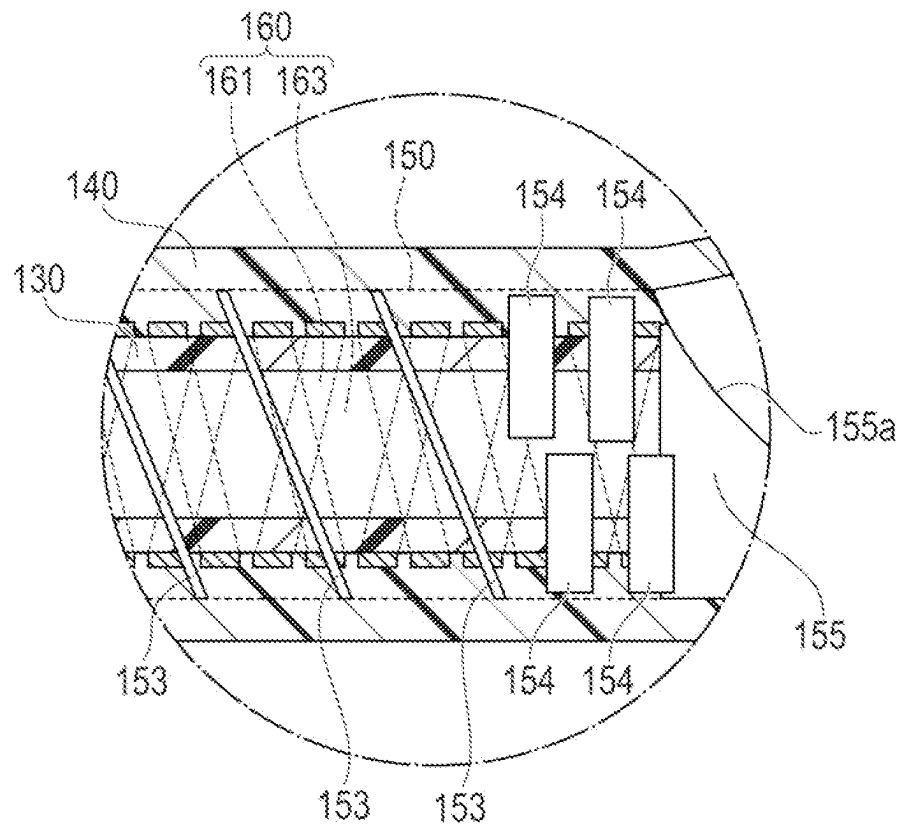


Fig. 7A

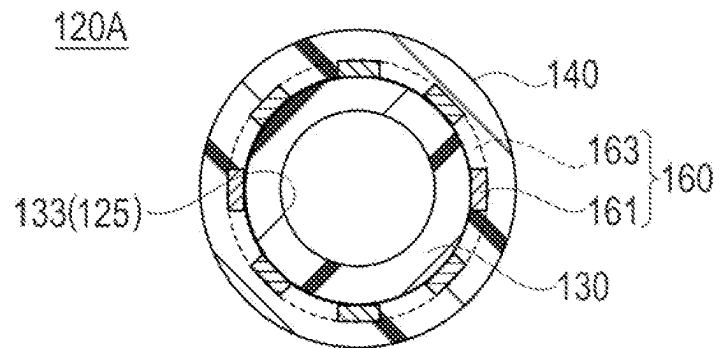


Fig. 7B

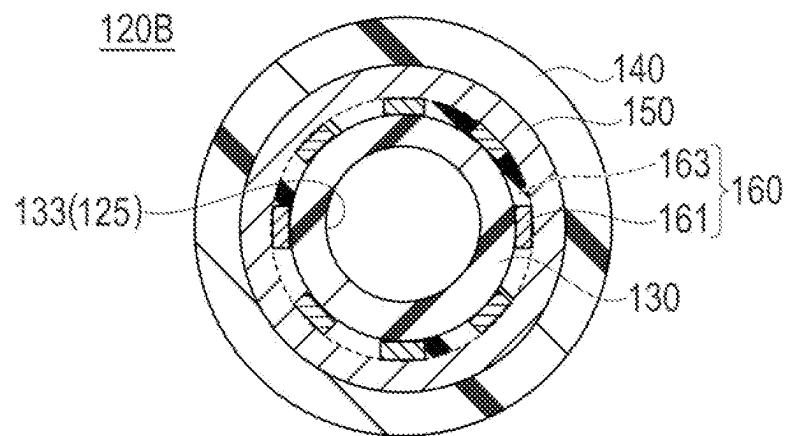


Fig. 7C

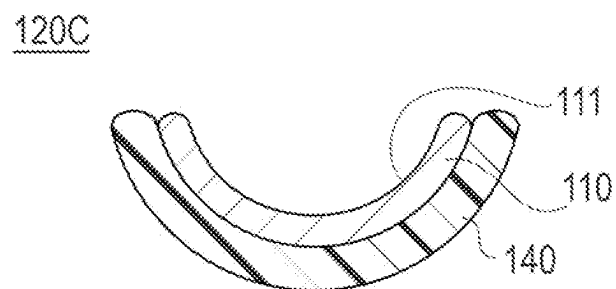


FIG. 8

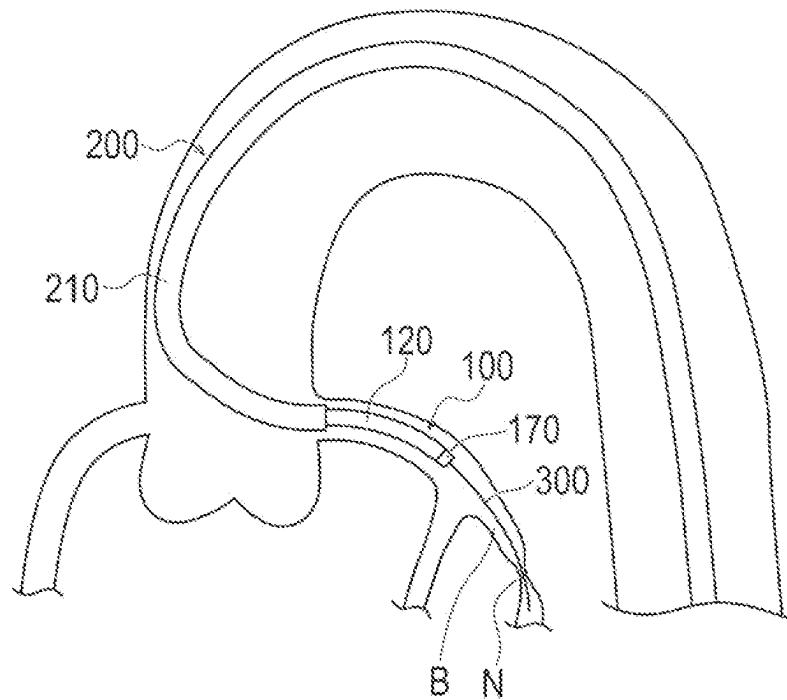
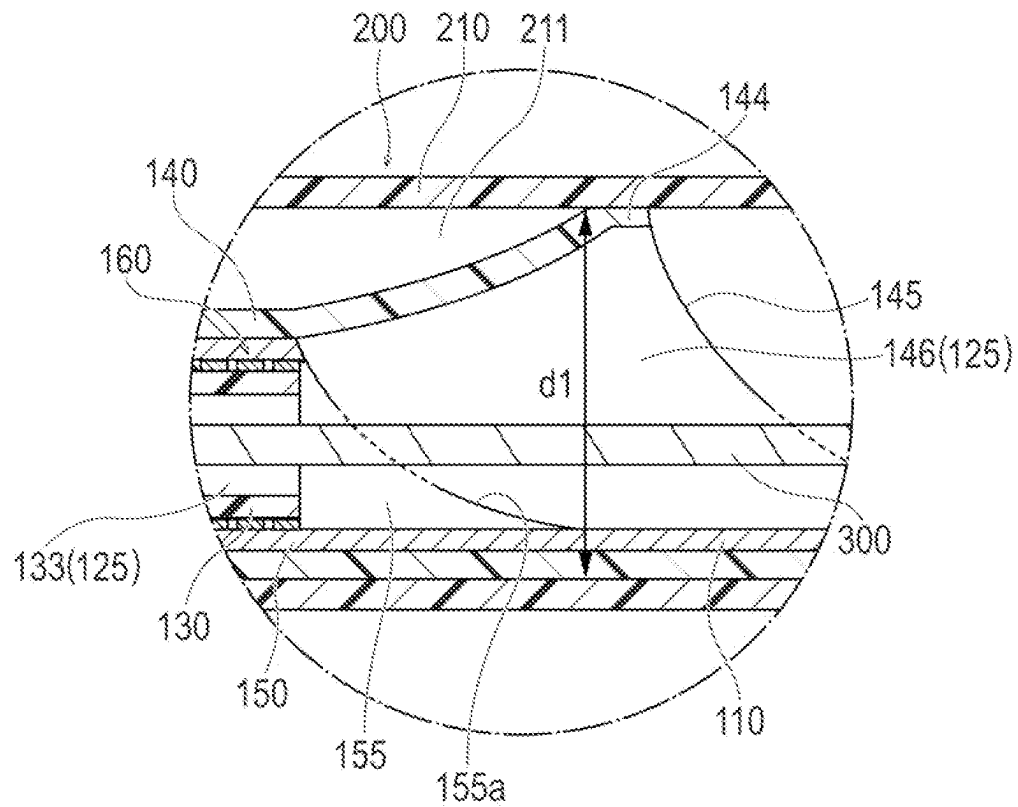


FIG. 9



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MEDICAL ELONGATED BODY AND MEDICAL INSTRUMENT SET

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/346,542 filed on Jun. 14, 2021, which is a continuation of U.S. application Ser. No. 16/583,411 filed on Sep. 26, 2019, now U.S. Pat. No. 11,134,982, which is a continuation of International Application No. PCT/JP2018/012430 filed on Mar. 27, 2018, which claims priority to Japanese Application No. 2017-071981 filed on Mar. 31, 2017, the entire contents of all four of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a medical elongated body and a medical instrument set.

BACKGROUND ART

When performing various procedures on a body lumen such as blood vessel, an operator delivers a guiding catheter (parent catheter) to a location close to a lesion area, for example, to an entrance portion of a coronary artery along a guide wire which has been inserted into the blood vessel in advance. After having guided a distal portion of the guiding catheter to the location close to the lesion area, for example, to the entrance portion of the coronary artery, the operator enables a treatment device such as balloon catheter to advance from the distal portion of the guiding catheter, and performs a procedure on the lesion area (for example, a stenosed site formed in the blood vessel).

For example, when the operator performs a procedure on the stenosed site formed in the coronary artery, if the stenosed site is spaced apart from the entrance of the coronary artery, since the treatment device receives various resistance due to influences of intravascular stenosis or the tortuousness of the blood vessel, it becomes relatively difficult for the operator to insert the treatment device to the lesion area. In this case, after having disposed the guiding catheter at a predetermined location inside the blood vessel, the operator inserts a medical elongated body (child catheter) into the guiding catheter, and delivers the medical elongated body to a position as close to the lesion area as possible. The operator is capable of relatively easily inserting the treatment device into the lesion area by supporting the delivery of the treatment device with the medical elongated body.

As an example of the medical elongated body described above, JP-A-2011-135989 discloses a rapid exchange type (monorail type) catheter including a tubular member (i.e., insertion tube) formed with a lumen into which a treatment device can be inserted, and a linear body (wire) provided on a proximal side of the tubular member.

The medical elongated body described in JP-A-2011-135989 has a covering tube overlapping an outer circumferential surface of a proximal portion of the tubular member, and a hook-shaped interlock portion formed by bending a distal portion of the linear body into a U shape. The hook-shaped interlock portion included in the linear body is connected to the tubular member via the covering tube in a state where the hook-shaped interlock portion is inserted into the lumen of the tubular member.

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In the rapid exchange type medical elongated body described in JP-A-2011-135989, the lumen is formed only in a tubular body disposed on a distal side, and thus the operator is capable of saving the time and effort taken to insert the treatment device over the entire length of the medical elongated body, and easily inserting the treatment device into the medical elongated body. In addition, the medical elongated body described in JP-A-2011-135989 has a small area of contact between a guiding catheter which is a parent catheter and the linear body of the medical elongated body. For this reason, the operator is capable of rather smoothly moving the medical elongated body along a lumen of the guiding catheter.

However, when the rapid exchange type medical elongated body described in JP-A-2011-135989 is used to support the delivery of the treatment device, problems can occur.

For example, since the tubular member and the linear body are separate members, the medical elongated body has a relatively large physical property difference in a connection section between the tubular member and the linear body. For this reason, when the medical elongated body is delivered to the vicinity of a lesion area, the medical elongated body has a potential for kinking in the connection section between the tubular member and the linear body.

In addition, in the rapid exchange type medical elongated body, it is necessary to prevent a proximal opening portion, which is provided on the proximal side of the tubular member, from moving to a distal side of the guiding catheter in the middle of a procedure using the medical elongated body. For this reason, in the middle of the procedure, it is necessary for the operator to figure out the position of the proximal side of the tubular member. However, in the medical elongated body, if a contrast marker (contrast ring) typically used in a catheter device is disposed in the proximal portion of the tubular member, the influence of the rigidity of the contrast marker increases a physical property difference in the vicinity of a boundary portion between the tubular member and the linear body. Therefore, the medical elongated body is likely to kink.

Note that in the medical elongated body described in JP-A-2011-135989, as described above, the hook-shaped interlock portion formed in the distal portion of the linear body overlaps the outer circumferential surface of the proximal portion of the tubular member. For this reason, if the linear body and the tubular member are formed of a metallic material, in the medical elongated body, there can be a partial improvement in the contrast property of the vicinity of the proximal portion of the tubular member. Since the range of a portion of the medical elongated body which has an improved contrast property is limited corresponding to an exterior shape (hook shape) of the linear body, it can be difficult to improve the contrast property of the vicinity of the proximal portion of the tubular member to a desired level.

In addition, in the medical elongated body, the outer diameter of the distal portion of the tubular member is preferably set smaller than the outer diameter of the proximal portion of the tubular member. Since the medical elongated body is formed as described above, the distal portion of the tubular member can be delivered relatively smoothly to the lesion area.

SUMMARY OF DISCLOSURE

A rapid exchange type medical elongated body is disclosed in which an outer diameter of a distal portion of a

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tubular member is smaller than an outer diameter of a proximal portion of the tubular member, which can help prevent an occurrence of a kink in a physical property change region of the medical elongated body which has a relatively large change in physical property, and in which a contrast property of the vicinity of the proximal portion of the tubular member can be improved, and a medical instrument set including the medical elongated body and a catheter.

According to an embodiment of the present disclosure, a medical elongated body is disclosed, which includes a shaft portion made of a linear body; and a distal member formed on a distal side of the shaft portion and having a lumen, in which the distal member has a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer, the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer, in which the tubular member forming the second region is integrally formed with the shaft portion, in which the distal member has a reinforcement layer containing a metallic material and formed between the inner layer and the outer layer from the first region to the second region, in which the reinforcement layer is disposed inside the tubular member in the second region, and in which an outer diameter of the first region is set smaller than an outer diameter of the second region.

In the medical elongated body with the foregoing configuration, the shaft portion made of a linear body is integrally formed with the tubular member forming the second region of the distal member. For this reason, in the medical elongated body, a physical property difference can be kept small at a boundary portion (physical property change region) between the shaft portion and the tubular member, and the occurrence of a kink can be prevented at the boundary portion.

In addition, the medical elongated body with the foregoing configuration is capable of improving the contrast property of the second region by virtue of the tubular member forming the second region, and the reinforcement layer containing a metallic material and disposed inside the tubular member in the second region. Therefore, it is not necessary to separately provide a contrast marker (contrast ring) for indicating a proximal side of the second region, and thus the medical elongated body is capable of improving the contrast property of the second region, and preventing an occurrence of a physical property difference resulting from the placement of the contrast marker.

In addition, since the outer diameter of the first region of the distal member is smaller than the outer diameter of the second region, the ease to insert the medical elongated body with the foregoing configuration into a body lumen can be improved.

In accordance with another aspect, a medical elongated body is disclosed comprising: a shaft portion; a distal member formed on a distal side of the shaft portion and having a lumen, the distal member having a first region, a second region, and a third region, the first region having an inner layer and an outer layer, the second region having the inner layer, the outer layer, and a tubular member, the tubular member being disposed between the inner layer and the outer layer in the second region, and an outer diameter of the first region being smaller than an outer diameter of the second region; a reinforcement layer containing a metallic material arranged between the inner layer and the outer layer from the first region to the second region of the distal member, the reinforcement layer being disposed inside of

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the tubular member in the second region; and the tubular member having an inclined region inclined toward the shaft portion in the third region, the third region including a proximal opening portion and the third region has an outer diameter increasing from a distal side toward a proximal side of the inclined region.

In accordance with an aspect, a medical instrument set is disclosed comprising: a medical elongated body which includes a shaft portion made of a linear body, a distal member formed on a distal side of the shaft portion and having a lumen, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer, the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the tubular member forming the second region being integrally formed with the shaft portion, a reinforcement layer containing a metallic material arranged between the inner layer and the outer layer from the first region to the second region of the distal member, the reinforcement layer being disposed inside of the tubular member in the second region, and wherein an outer diameter of the first region is smaller than an outer diameter of the second region; and a catheter having a catheter main body portion with a lumen into which the medical elongated body is insertable, and wherein a maximum outer diameter of the proximal opening portion of the medical elongated body is equal to or greater than an inner diameter of the catheter main body portion.

In accordance with another aspect, a medical elongated body is disclosed comprising: a shaft portion made of a linear body comprising a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal opening portion and a proximal opening portion, the distal member including a lumen extending from the distal opening portion to the proximal opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer, the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the tubular member forming the second region being integrally formed with the shaft portion, the proximal opening portion being disposed at a proximal end of the lumen; a reinforcement layer containing a metallic material arranged between the inner layer and the outer layer of the first region to the inner layer and the outer layer of the second region of the distal member, the reinforcement layer being disposed inside of the tubular member in the second region; and wherein an outer diameter of the first region is smaller than an outer diameter of the second region.

In accordance with an aspect, a medical elongated body is disclosed comprising: a linear body shaft portion comprising a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal opening portion and a proximal opening portion, the distal member including a lumen extending from the distal opening portion to the proximal opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer, the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the tubular member forming the second region being integrally formed with the linear body shaft portion, the proximal opening portion being disposed at a proximal end of the lumen of the distal member; a reinforcement layer containing a metallic mate-

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rial arranged between the inner layer and the outer layer of the first region to the inner layer and the outer layer of the second region of the distal member, the reinforcement layer being disposed inside of the tubular member in the second region; and wherein an outer diameter of the first region is smaller than an outer diameter of the second region.

In accordance with another aspect, a medical elongated body is disclosed comprising: a linear body shaft portion comprising a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal opening portion and a proximal opening portion, the distal member includes a lumen extending from the distal opening portion to the proximal opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer, the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the tubular member forming the second region being integrally formed with the shaft portion, the proximal opening portion being disposed at a proximal end of the lumen, and the proximal opening portion having an inclined region toward the linear body shaft portion; a reinforcement layer containing a metallic material arranged between the inner layer and the outer layer of the first region to the inner layer and the outer layer of the second region of the distal member, the reinforcement layer being disposed inside of the tubular member in the second region; and wherein an outer diameter of the first region is smaller than an outer diameter of the second region.

In accordance with an aspect, a medical elongated body comprising: a shaft portion comprising a linear body and a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer and the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the proximal end opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer from the first region to the second region of the distal member.

In accordance with a further aspect, a medical elongated body comprising: a shaft portion comprising a linear body and a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer and the outer layer, and a tubular member disposed between the inner layer and the outer layer in the

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second region, the proximal end opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer of the first region of the distal member.

In accordance with an aspect, a medical elongated body comprising: a shaft portion comprising a linear body and a proximal portion; a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a first region resin material, and a second region having an inner layer and an outer layer which are made of a second region resin material, and a tubular member disposed between the inner layer and the outer layer in the second region, the tubular member has an opening portion, the outer layer of the second region is welded to the inner layer via the opening portion, the proximal end opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer of the first region.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a simplified view illustrating a medical instrument set according to an embodiment of the present disclosure.

FIG. 2 is a perspective view of a medical elongated body according to the embodiment.

FIG. 3 is an enlarged cross-sectional view of a distal member of the medical elongated body along an axial direction.

FIG. 4 is an enlarged cross-sectional view illustrating a second region and a third region of the distal member of the medical elongated body.

FIG. 5 is an enlarged view illustrating a portion surrounded by a broken line 5A of FIG. 3.

FIG. 6 is an enlarged view illustrating a portion surrounded by a broken line 6A of FIG. 4.

FIG. 7A is an axially orthogonal cross-sectional view of a first region of the distal member taken along an arrow-head line 7A-7A of FIG. 3.

FIG. 7B is an axially orthogonal cross-sectional view of the second region of the distal member taken along an arrow-head line 7B-7B of FIG. 3.

FIG. 7C is an axially orthogonal cross-sectional view of the third region of the distal member taken along an arrow-head line 7C-7C of FIG. 3.

FIG. 8 is a schematic view illustrating a use example of the medical instrument set according to the embodiment.

FIG. 9 is an enlarged cross-sectional view for describing effects of the medical instrument set according to the embodiment.

DESCRIPTION OF EMBODIMENTS

Set forth below with reference to the accompanying drawings is a detailed description of embodiments of a medical elongated body and a medical instrument set representing examples of the inventive medical elongated body and medical instrument disclosed here. The dimensions or scales on the drawings may be exaggerated or different from actuality/reality for convenience of description and illustration.

Medical Instrument Set

FIG. 1 illustrates a medical instrument set (catheter set) 10 according to the embodiment.

As illustrated in FIG. 1, the medical instrument set 10 has a medical elongated body 100 and a catheter 200.

In accordance with an exemplary embodiment, the catheter 200 can be configured as a so-called guiding catheter that delivers a medical device (for example, a treatment device such as balloon catheter or stent delivery device, or a diagnosis device such as intravascular ultrasound diagnosis catheter or intravascular optical diagnosis catheter) to a lesion area such as stenosed site when a body lumen such as blood vessel is treated or diagnosed. Generally, the catheter 200 can also be referred to as a parent catheter.

In accordance with an exemplary embodiment, the medical elongated body 100 can be configured as a so-called support catheter that is used for a procedure such as inserting the medical device, which has been delivered to the vicinity of the lesion area such as stenosed site by the catheter 200, into the lesion area, or such as causing the medical device to pass through the lesion area. Generally, the medical elongated body 100 is also referred to as a child catheter.

Catheter

As illustrated in FIG. 1, the catheter 200 has a catheter main body portion 210 formed with a lumen 211 into which a guide wire, the medical device, the medical elongated body 100, or the like can be inserted, and a hub 220 disposed in a proximal portion of the catheter main body portion 210.

In accordance with an exemplary embodiment, the catheter main body portion 210 can be a hollow tubular member having relative flexibility. A distal opening portion 211a is formed at a distal end of the catheter main body portion 210, and communicates with the lumen 211 of the catheter main body portion 210. The catheter main body portion 210 can be formed of (i.e., made of), for example, a resin material.

In accordance with an exemplary embodiment, the hub 220 of the catheter 200 has a Luer tapered connector (Y-connector) 221 that is known in the medical field. The connector 221 has a side branch 223 connected to a tubular member through which a contrast agent or a priming liquid can flow; a valve body 224 disposed inside the hub 220; and an opener 225 for opening and closing the valve body 224. In addition, a proximal port (not illustrated) configured to receive a shaft portion 110 or a distal member 120 of the medical elongated body 100 into and out from the hub 220 can be provided on a proximal side of the hub 220.

When performing a procedure using the medical elongated body 100, an operator inserts a distal side (the shaft portion 110 and the distal member 120) of the medical elongated body 100 into the hub 220 and the catheter main body portion 210 via the proximal port provided in the hub 220 of the catheter 200. If the operator supplies the contrast agent from the side branch 223 in a state where the distal side of the medical elongated body 100 is inserted into the catheter main body portion 210 of the catheter 200, the contrast agent can be discharged to a distal side of the catheter main body portion 210. At that time, the operator is

capable of preventing the contrast agent from flowing out from the proximal port of the hub 220 by performing a closing operation of the valve body 224 of the hub 220.

Medical Elongated Body

As illustrated in FIGS. 1 and 2, the medical elongated body 100 has the shaft portion 110 made of a linear body; the distal member 120 formed on a distal side of the shaft portion 110; and a grip portion 180 which is disposed on a proximal side of the shaft portion 110 and can be gripped by the fingers of the operator.

As illustrated in FIGS. 1 and 2, the distal member 120 has a first region 120A, a second region 120B, and a third region 120C which are disposed in sequence from a distal side to a proximal side. Note that in FIG. 2, the member number of the main member forming the regions is assigned to each of the regions 120A, 120B, and 120C in order to schematically describe the structure of the distal member 120.

As illustrated in FIGS. 3 and 7A, the first region 120A of the distal member 120 has an inner layer 130 made of a resin material, and an outer layer 140 made of a resin material.

Each of the inner layer 130 and the outer layer 140 is formed as a hollow tubular member extending in an axial direction (right and left direction in FIG. 3) of the distal member 120.

As illustrated in FIG. 3, the inner layer 130 has a lumen 133 formed inside of the inner layer 130. The lumen 133 of the inner layer 130 communicates with a lumen 146 of a proximal portion (a portion of the outer layer 140, which is disposed in the third region 120C) 144 of the outer layer 140. The lumen 133 of the inner layer 130 and the lumen 146 of the proximal portion 144 of the outer layer 140 form a lumen 125 through which the guide wire, the medical device, or the like is inserted into the distal member 120.

As illustrated in FIG. 3, a distal tip 170 is provided on the distal side of the distal member 120, and helps prevent a distal end of the medical elongated body 100 from causing damages to a body organ when the distal end comes into contact with a body lumen (intravascular wall or the like).

The distal tip 170 has a distal opening portion 171 opening at a distal end of the distal tip 170. In addition, the distal tip 170 is fixed (for example, welded) to the inner layer 130 in a state where the distal tip 170 covers a portion of the inner layer 130, which is not covered with the outer layer 140.

As illustrated in FIG. 5, the outer layer 140 of the distal member 120 can have an inclined portion 143 having an outer diameter decreasing from a distal end of the second region 120B toward the distal side. In accordance with an exemplary embodiment, the inclined portion 143 has a cross-sectional shape that is curved (i.e., gently curved) from a proximal side to the distal side.

The outer diameter (an outer diameter D1 in the first region 120A) of the outer layer 140 in the first region 120A can be, for example, from 1.4 mm to 1.8 mm. In addition, the outer diameter (an outer diameter D2 in the second region 120B) of the outer layer 140 in the second region 120B can be, for example, from 1.7 mm to 2.1 mm.

As illustrated in FIG. 5, in an axially orthogonal cross section of the distal member 120, a wall thickness t1 of the outer layer 140 in the first region 120A closer to the distal side than the inclined portion 143 is substantially the same as a wall thickness t2 of the outer layer 140 in the second region 120B. Note that the range of substantially the same dimension contains, for example, a dimensional difference induced by manufacturing tolerances.

In accordance with an exemplary embodiment, a maximum wall thickness t3 of the inclined portion 143 can be

greater than the wall thickness **t1** in the first region **120A** and the wall thickness **t2** in the second region **120B**. Transition in the wall thickness of the inclined portion **143** is made such that the wall thickness becomes maximum (i.e., greatest wall thickness) in the vicinity of the distal end of the second region **120B** and becomes minimum (i.e., least wall thickness) in the vicinity of a proximal end of the first region **120A**.

In accordance with an exemplary embodiment, the outer layer **140** can be formed, for example, by joining together a resin tube (i.e., a tubular member formed from a resin) forming the first region **120A** and a resin tube forming the second region **120B**. In this case, for example, the tubes (i.e., tubular members) can be welded together by applying heat to the tubes in a state where the resin tube forming the first region **120A** is covered with the resin tube forming the second region **120B**. In addition, at that time, the wall thickness of an overlapping portion of both tubes (i.e., tubular members) increases by the difference in outer diameter between both tubular members. In accordance with an exemplary embodiment, the first region **120A**, the inclined portion **143**, and the second region **120B** can be integrally formed, each of which includes a predetermined wall thickness, by adjusting the portion with an increased wall thickness such that the outer diameter of the portion decreases to the distal side as illustrated in FIG. 5.

The wall thickness **t1** of the first region **120A** can be, for example, from 0.04 mm to 0.25 mm. In addition, the wall thickness **t2** of the second region **120B** can be, for example, from 0.10 mm to 0.30 mm. In addition, the maximum wall thickness **t3** of the inclined portion **143** can be, for example, from 0.04 mm to 0.40 mm.

As illustrated in FIGS. 3 and 7B, the second region **120B** of the distal member **120** has the inner layer **130**; the outer layer **140**; and a tubular member **150** made of a metallic material and disposed between the inner layer **130** and the outer layer **140**.

In accordance with an exemplary embodiment, the tubular member **150** forming the second region **120B** can be integrally formed with the shaft portion **110** extending to a proximal side of the tubular member **150**.

As illustrated in FIGS. 4 and 7B, the tubular member **150** has a hollow tubular structure. In the third region **120C** formed closer to the proximal side than the second region **120B**, the tubular member **150** has an inclined region **155** inclined toward the shaft portion **110**. A proximal opening portion **155a** opening to the proximal side is formed in the inclined region **155**.

In accordance with an exemplary embodiment, as illustrated in FIG. 4, the tubular member **150** is connected to and integrated with the shaft portion **110**, in a region closer to the proximal side than the inclined region **155**.

As illustrated in FIGS. 4 and 7C, the distal side of the shaft portion **110** is formed into substantially a semicircular arc shape which is the shape of an axially orthogonal cross section. In accordance with an exemplary embodiment, an inner surface **111** of the shaft portion **110** has a concavely curved cross-sectional shape. Since the shaft portion **110** is formed into substantially a semicircular arc shape as described above, when the shaft portion **110** is inserted into the catheter main body portion **210** of the catheter **200**, the shaft portion **110** is capable of reducing a sliding resistance applied between the catheter main body portion **210** and the shaft portion **110**.

As illustrated in FIG. 2, the shaft portion **110** extends in substantially a semicircular arc shape to the proximal side on

which the grip portion **180** is disposed. The grip portion **180** is fixed to the proximal side of the shaft portion **110**.

As illustrated in FIGS. 3, 7A and 7B, the distal member **120** has a reinforcement layer **160** containing a metallic material and formed between the inner layer **130** and the outer layer **140** from the first region **120A** to the second region **120B**.

As illustrated in FIG. 7B, the reinforcement layer **160** is disposed inside the tubular member **150** in the second region **120B** of the distal member **120**.

As illustrated in FIG. 6, the reinforcement layer **160** has a reinforcement member **161** formed of a metallic material, and a gap portion **163** formed in the reinforcement member **161**.

In accordance with an exemplary embodiment, the gap portion **163** of the reinforcement member **161** connects together the inner layer **130** and the outer layer **140** in a state where the gap portion **163** is disposed between the inner layer **130** and the outer layer **140**. Note that the connection implies a state where the inner layer **130** and the outer layer **140** are in physical contact with each other or a state where the inner layer **130** and the outer layer **140** can be, for example, melted and solidified together.

In accordance with an exemplary embodiment, the reinforcement member **161** of the reinforcement layer **160** is formed of a metallic material. The reinforcement layer **160** has a tubular structure formed by braiding a wire rod formed of a metallic material. A braided wire-rod portion forms the reinforcement members **161**, and a mesh portion shaped by the reinforcement members **161** forms the gap portions **163**.

The cross-sectional shape of the reinforcement member **161** is not specifically limited, and can be, for example, a rectangular or circular shape. In addition, for example, a one over and one under pattern in which the reinforcement members **161** intersect each other can be adopted as a braid pattern of the reinforcement member **161**.

Note that the reinforcement layer **160** may be formed as, for example, a coil helically shaped along an axial direction.

As illustrated in FIG. 4, the tubular member **150** has a slit portion **153**.

As illustrated in FIGS. 4 and 7B, in the second region **120B**, the outer layer **140** can be welded to the inner layer **130** via each gap formed by the gap portion **163** of the reinforcement layer **160** and the slit portion **153** of the tubular member **150**.

In order to help enable the inner layer **130** and the outer layer **140** to be welded together via both of the gap portion **163** and the slit portion **153**, at least parts of both can be disposed to overlap each other in the axial direction and a circumferential direction. In accordance with an exemplary embodiment, the foregoing gap is formed in a region where the gap portion **163** and the slit portion **153** overlap each other in the axial direction and the circumferential direction.

As illustrated in FIG. 4, the slit portion **153** of the tubular member **150** is helically formed in the tubular member **150**. In addition, a pitch **P** (axial distance between the slit portions **153** adjacent to each other) between the slit portions **153** can be such that the pitch **P** decreases from a proximal end toward a distal end of the tubular member **150**.

As illustrated in FIG. 6, the tubular member **150** has a plurality of opening portions **154** in a region closer to a proximal side than a region where the slit portion **153** is formed. Therefore, the outer layer **140** can be reliably welded to the inner layer **130** via the opening portions **154**, in a proximal portion of the tubular member **150** which is likely to come into contact with the medical device inserted into the distal member **120**. Note that each of the slit portion

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153 and the opening portion 154 is a hole (space) which is actually a non-existing member, but in FIG. 6, in order to clearly illustrate a positional relationship between both, the slit portion 153 and the opening portion 154 are illustrated in white.

The opening portion 154 is a hole that passes through the tubular member 150 in the wall thickness direction. A specific shape or the number of the opening portions 154 is not specifically limited.

Note that if the pitch P of the slit portion 153 of the tubular member 150 is such that the pitch P decreases from the proximal end toward the distal end of the tubular member 150, since the tubular member 150 has a small number of the slit portions 153 on the proximal side of the tubular member 150, it can be difficult for the gap portion 163 of the reinforcement layer 160 and the slit portion 153 of the tubular member 150 to form a gap, and it can be difficult to weld together the inner layer 130 and the outer layer 140. In this case, since the tubular member 150 has the opening portion 154 provided closer to the proximal side than the region where the slit portion 153 is formed, the inner layer 130 and the outer layer 140 can be suitably welded together.

As illustrated in FIG. 5, a distal portion 151 of the tubular member 150 can be formed into a tapered shape, the outer diameter of which decreases to the proximal side. Note that the shape of an end surface of the distal portion 151 of the tubular member 150 is not limited to a shape (rounded shape) which is curved (i.e., gently curved) to the distal side as illustrated, and for example, a distal surface may have a straight line shape which perpendicularly intersects the axial direction.

In addition, the tubular member 150 may have a shape in which the inner diameter and the outer diameter of the tubular member 150 decreases to the distal side in a stepwise manner or decreases continuously. It is possible to form the tubular member 150 with the foregoing shape, for example, the shape is formed by fixing the tubular member 150 to the reinforcement layer 160 with a swage process in a state where the reinforcement layer 160 is covered with the tubular member 150.

As illustrated in FIG. 4, the third region 120C of the distal member 120 is formed with a proximal opening portion 145 which is formed of a material more flexible than the material of the tubular member 150 and has an outer diameter increasing from a distal side toward a proximal side of the inclined region 155 of the tubular member 150.

The proximal opening portion 145 formed in the third region 120C is formed on a proximal side of the outer layer 140. In accordance with an exemplary embodiment, the lumen 146 of the proximal portion 144 of the outer layer 140 has a shape which becomes gradually wider (i.e., larger) to the proximal side, and the proximal opening portion 145 is formed at a proximal end of the lumen 146.

As illustrated in FIG. 4, the tubular member 150 has a cylindrical tubular structure in the vicinity of a distal end of the inclined region 155 of the tubular member 150. In a cylindrical portion of the tubular member 150, the tubular member 150 and the reinforcement layer 160 disposed inside the tubular member 150 overlap each other in the circumferential direction. In accordance with an exemplary embodiment, the shape of the tubular member 150 can be gradually transformed from the cylindrical shape to the shape (semicircular arc shape) of the shaft portion 110 from the distal side to the proximal side of the inclined region 155.

In accordance with an exemplary embodiment, a maximum diameter D3 of the proximal opening portion 145 formed in the third region 120C can be, for example,

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substantially equal to or greater than an inner diameter (diameter of the lumen 211) d1 (refer to FIG. 9) of the catheter main body portion 210 of the catheter 200. Note that the range of substantially the same dimension, for example, can contain a dimensional difference induced by manufacturing tolerances.

In accordance with an exemplary embodiment, if the inner diameter d1 of the catheter main body portion 210 of the catheter 200, for example, is from 1.4 mm to 2.2 mm, the maximum diameter D3 of the proximal opening portion 145, for example, is preferably from 1.4 mm to 2.2 mm, and more preferably from 1.7 mm to 1.9 mm.

If it is possible to cause the medical elongated body 100 to protrude from the catheter main body portion 210 by a desired length when using the medical elongated body 100 along with the catheter 200, the length of a portion of the medical elongated body 100 along the axial direction, which contains the distal member 120 and the shaft portion 110, is not specifically limited, and can be, for example, from 800 mm to 1,300 mm. In addition, the length of the distal member 120 along the axial direction can be, for example, from 100 mm to 300 mm. In addition, the length of the second region 120B of the distal member 120 along the axial direction can be, for example, from 10 mm to 100 mm. In addition, the length of the third region 120C of the distal member 120 along the axial direction can be, for example, from 5 mm to 50 mm.

Subsequently, a configuration material of the medical elongated body 100 will be described.

In accordance with an exemplary embodiment, the inner layer 130 and the outer layer 140 of the distal member 120 are each preferably made of a resin material more flexible than the material of the tubular member 150. The inner layer 130 and the outer layer 140 can be made of, for example, a polyolefin such as polyethylene, polypropylene, an ethylene-propylene copolymer, or an ethylene-vinyl acetate copolymer, a thermoplastic resin such as soft polyvinyl chloride, any of various elastomers such as polyurethane elastomer, polyamide elastomer, or polyester elastomer, fluorine resin such as polytetrafluoroethylene, polyamide, or a crystalline plastic such as crystalline polyethylene or crystalline polypropylene.

The distal tip 170 can be made from, for example, a polymer material such as polyolefin (for example, polyethylene, polypropylene, polybutene, ethylene-propylene copolymer, ethylene-vinyl acetate copolymer, ionomer, or a mixture of two or more of the polymer materials), polyvinyl chloride, polyamide, polyamide elastomer, polyurethane, polyurethane elastomer, polyimide, or fluorine resin, any mixture of the polymer materials, or a multilayer tubular member made of two or more of the polymer materials. Note that the distal tip 170 is preferably made from a material more flexible than the materials of the inner layer 130 and the outer layer 140.

In accordance with an exemplary embodiment, the shaft portion 110 and the tubular member 150 can be made from a metallic material, such as stainless steel or super-elastic alloy such as Ni—Ti-based alloy, Ni—Al-based alloy, or Cu—Zn-based alloy.

For example, the reinforcement member 161 of the reinforcement layer 160 can be made from the same materials as the exemplary materials of the shaft portion 110 and the tubular member 150.

Subsequently, a use example of the medical instrument set 10 (the medical elongated body 100 and the catheter 200) will be described.

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For example, when performing a procedure on a lesion area N such as stenosed site formed in a coronary artery by using a medical device (not illustrated) such as balloon catheter or stent delivery device, an operator or the like disposes a distal end of the catheter **200** in the vicinity of an entrance of the coronary artery so that the medical device inserted into the catheter **200** can be supported when being delivered to the lesion area. However, if the lesion area N such as stenosed site is spaced apart from the vicinity of the entrance of the coronary artery, even though the distal end of the catheter **200** is disposed in the vicinity of the entrance of the coronary artery, the medical device may not receive a sufficient support force when being inserted into the catheter **200**. In this case, the operator inserts the catheter **200** further inside (to the lesion area N) of the coronary artery, and attempts to deliver the medical device to the lesion area N in a state where the medical device has received greater support force.

At that time, if the catheter **200** has a relatively large diameter, the catheter **200** has a potential for causing ischemia or the like. For this reason, as illustrated in FIG. **8**, the operator delivers the medical elongated body **100**, which is thinner, to a distal side of the catheter **200** via the catheter **200**. Specifically, if the lesion area N such as stenosed site is spaced apart from the vicinity of the entrance of the coronary artery, before inserting the medical device or after extracting the medical device from the catheter **200** for the moment, the operator inserts the medical elongated body **100**.

As illustrated in FIG. **9**, when delivering the medical device, the operator delivers the medical elongated body **100** to a desired position along a guide wire **300** inserted into the catheter **200**. The operator can cause the distal side of the distal member **120** of the medical elongated body **100** to protrude from the catheter **200** by a predetermined length, and delivers the medical device via the medical elongated body **100** in a state where the medical device has received a desired support force.

When the medical elongated body **100** is delivered via the catheter **200**, as illustrated in FIG. **9**, the proximal portion **144** of the outer layer **140** is brought into contact with an inner surface of the catheter main body portion **210** of the catheter **200**. At that time, the proximal portion **144** of the outer layer **140** can be flexibly deformed along the inner surface of the catheter main body portion **210**, and a clearance between the inner surface of the catheter main body portion **210** and the proximal portion **144** decreases. Therefore, the operator can be capable of rather smoothly moving the medical elongated body **100** along the catheter **200**. In addition, the operator can prevent the medical device, which is inserted into the lumen **125** of the distal member **120** from the proximal opening portion **145** of the outer layer **140**, from being caught in the clearance by decreasing the clearance between the proximal portion **144** of the outer layer **140** and the inner surface of the catheter main body portion **210**. Therefore, the operator can rather smoothly insert the medical device into the lumen **125** of the distal member **120**.

Furthermore, when supplying the contrast agent from the side branch **223** of the catheter **200**, the operator can help prevent an occurrence of an event such as the contrast agent flowing into the clearance between the proximal portion **144** of the outer layer **140** and the inner surface of the catheter main body portion **210** and being charged in full from the distal opening portion **211a** of the catheter main body portion **210** (refer to FIG. **1**). For example, the operator can cause the contrast agent to be sent to the distal side of the

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medical elongated body **100** which has protruded to the distal side of the catheter **200**, and the contrast agent to be discharged to the lesion area which is a treatment target, and an area in the periphery of the lesion area. Therefore, the operator can acquire a desired contrast image.

Subsequently, effects of the medical elongated body **100** and the medical instrument set **10** according to the embodiment will be described.

The medical elongated body **100** according to the present disclosure has the shaft portion **110** made of a linear body, and the distal member **120** which is formed on the distal side of the shaft portion **110** and has the lumen **125**. The distal member **120** has the first region **120A** having the inner layer **130** and the outer layer **140** which are made of a resin material, and the second region **120B** having the inner layer **130**, the outer layer **140**, and the tubular member **150** which is made of a metallic material and is disposed between the inner layer **130** and the outer layer **140**. The tubular member **150** forming the second region **120B** is integrally formed with the shaft portion **110**. The distal member **120** has the reinforcement layer **160** containing a metallic material and formed between the inner layer **130** and the outer layer **140** from the first region **120A** to the second region **120B**. Furthermore, the reinforcement layer **160** is disposed inside the tubular member **150** in the second region **120B**. The outer diameter of the first region **120A** is set smaller than the outer diameter of the second region **120B**.

In the medical elongated body **100** with the foregoing configuration, the shaft portion **110** made of a linear body can be integrally formed with the tubular member **150** forming the second region **120B** of the distal member **120**. For this reason, in the medical elongated body **100**, a physical property difference in a boundary portion (physical property change region) between the shaft portion **110** and the tubular member **150** can be kept relatively small, and an occurrence of a kink can be prevented.

In addition, the medical elongated body **100** with the foregoing configuration is capable of improving the contrast property of the second region **120B** by virtue of the tubular member **150** forming the second region **120B**, and the reinforcement layer **160** containing a metallic material and disposed inside the tubular member **150** in the second region **120B**. For this reason, it is not necessary to separately provide a contrast marker (contrast ring) for indicating the vicinity of a proximal portion of the second region **120B**, and thus the medical elongated body **100** with the configuration as disclosed is capable of improving the contrast property of the second region **120B**, and preventing an occurrence of a physical property difference which results from the placement of the contrast marker.

In addition, since the outer diameter of the first region **120A** of the distal member **120** is smaller than the outer diameter of the second region **120B**, the ease to insert the medical elongated body **100** with the foregoing configuration into a body lumen can be improved.

In accordance with an exemplary embodiment, the outer layer **140** of the distal member **120** according to the embodiment has the inclined portion **143**, the outer diameter of which decreases from the distal end of the second region **120B** toward the distal side. For this reason, it is possible to prevent a deterioration in the ease to insert the medical elongated body **100** which is caused due to the second region **120B** being caught by the catheter **200**, the body lumen, or the like. Therefore, when inserting the medical elongated body **100** into the body lumen, an operator can smoothly insert the first region **120A** and consecutively the second region **120B** into the body lumen.

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In addition, in the medical elongated body **100** according to the embodiment, in the axially orthogonal cross section of the distal member **120**, the wall thickness **t1** of the outer layer **140** in the first region **120A** closer to the distal side than the inclined portion **143** is substantially the same as the wall thickness of the outer layer **140** in the second region **120B**. For this reason, the physical property difference in the boundary portion between the first region **120A** and the second region **120B** can be kept relatively small, and an occurrence of a kink can be prevented.

In addition, the reinforcement layer **160** according to the embodiment has the reinforcement member **161** formed of a metallic material, and the gap portion **163** which is formed in the reinforcement member **161** and connects together the inner layer **130** and the outer layer **140**. In addition, the tubular member **150** has the slit portion **153**. In addition, in the second region **120B**, the outer layer **140** is welded to the inner layer **130** via each gap formed by the gap portion **163** of the reinforcement layer **160** and the slit portion **153** of the tubular member **150**. For this reason, in the medical elongated body **100**, it is possible to suitably weld together the inner layer **130** and the outer layer **140** which are disposed such that the reinforcement member **161** formed of a metallic material and the tubular member **150** formed of a metallic material can be interposed (i.e., placed) between inner layer **130** and the outer layer **140**.

In addition, the slit portion **153** of the tubular member **150** according to the embodiment is helically formed in the tubular member **150**. The pitch of the slit portion **153** decreases from the proximal end toward the distal end of the tubular member **150**. For this reason, in the medical elongated body **100**, the flexibility of a distal side (the distal side of the tubular member **150**) of the second region **120B** improves, and thus the ease to insert the second region **120B** into the body lumen improves. Furthermore, since the tubular member **150** has the plurality of opening portions **154** in the region closer to the proximal side than the region where the slit portion **153** is formed in the tubular member **150**, it is possible to suitably weld together the inner layer **130** and the outer layer **140** on the proximal side of the tubular member **150** via the opening portions **154**.

In addition, the distal member **120** according to the embodiment has the third region **120C** closer to the proximal side than the second region **120B**. In the third region **120C**, the tubular member **150** has the inclined region **155** inclined toward the shaft portion **110**. The third region **120C** is formed with the proximal opening portion **145** which is formed of a material more flexible than the material of the tubular member **150**, and the outer diameter of which increases from the distal side toward the proximal side of the inclined region **155**. For this reason, when the medical elongated body **100** is inserted into the catheter main body portion **210** of the catheter **200**, it is possible to decrease the clearance between the medical elongated body **100** and the catheter main body portion **210**, and a movement of the medical elongated body **100** with respect to the catheter **200** becomes relatively smooth. Furthermore, the medical elongated body **100** is capable of improving the ease of inserting the medical device into the lumen **125** of the medical elongated body **100**, and discharging the contrast agent to a proper position inside the body lumen.

In addition, the medical instrument set **10** according to the embodiment can include the medical elongated body **100**, and the catheter **200** having the catheter main body portion **210** formed with the lumen **211** into which the medical elongated body **100** can be inserted. In accordance with an exemplary embodiment, the maximum outer diameter of the

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proximal opening portion **145** of the medical elongated body **100** can be equal to or greater than the inner diameter of the catheter main body portion **210**. For this reason, in a procedure using the medical elongated body **100** and the catheter **200**, the medical instrument set **10** can smoothly deliver the medical elongated body **100** and the medical device to a desired position inside the body lumen, and properly discharging the contrast agent to the desired position inside the body lumen.

The medical elongated body according to the present disclosure has been described with reference to the embodiment, but the present disclosure is not limited only to the configuration described in the embodiment, and can be appropriately changed based on description of the claims.

For example, the distal member of the medical elongated body may have at least the first region and the second region, and may not have the third region.

In addition, the cross-sectional shape of the shaft portion of the medical elongated body is not limited to the shape (semicircular arc shape) illustrated, and can be appropriately changed.

In addition, the catheter described in the embodiment is merely an example, and if the medical elongated body can be inserted into and delivered by the catheter, the configuration of the catheter is not specifically limited.

In addition, each configuration part of the medical elongated body can be replaced with a desired part capable of exhibiting the same function. In addition, a desired component (member) which has not been specifically described in the specification may appropriately be added to the medical elongated body.

The detailed description above describes to a medical elongated body and a medical instrument set. The invention is not limited, however, to the precise embodiments and variations described. Various changes, modifications and equivalents can be effected by one skilled in the art without departing from the spirit and scope of the invention as defined in the accompanying claims. It is expressly intended that all such changes, modifications and equivalents which fall within the scope of the claims are embraced by the claims.

What is claimed is:

1. A medical elongated body comprising:

- a shaft portion comprising a linear body and a proximal portion;
- a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer and the outer layer, and a tubular member made of a metallic material and disposed between the inner layer and the outer layer in the second region, the proximal end opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and

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a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer from the first region to the second region of the distal member.

2. The medical elongated body according to claim 1, wherein the tubular member includes a slit portion, and wherein the outer layer of the second region is welded to the inner layer via the slit portion.

3. The medical elongated body according to claim 1, wherein the tubular member has an opening portion in a side wall and wherein the outer layer of the second region is welded to the inner layer via the opening portion.

4. The medical elongated body according to claim 1, wherein an outer diameter of the first region is smaller than an outer diameter of the second region.

5. The medical elongated body according to claim 1, wherein the outer layer of the distal member has an inclined portion having an outer diameter decreasing from a distal end of the second region toward a distal side of the distal member.

6. The medical elongated body according to claim 5, wherein in an axially orthogonal cross section of the distal member, a wall thickness of the outer layer in the first region closer to the distal side than the inclined portion is the same as a wall thickness of the outer layer in the second region.

7. The medical elongated body according to claim 2, wherein the slit portion is helically formed in the tubular member, and a pitch of the slit portion decreases from a proximal end toward a distal end of the tubular member.

8. The medical elongated body according to claim 2, wherein the tubular member includes a slit portion, the opening portion in a region being closer to a proximal side than a region where the slit portion is formed.

9. The medical elongated body according to claim 1, wherein the distal member has a third region, the third region being closer to a proximal side of the distal member than the second region.

10. The medical elongated body according to claim 9, wherein the third region has an outer diameter increasing from a distal side toward a proximal side of the inclined region of the proximal end opening portion.

11. The medical elongated body according to claim 9, wherein a material of the third region is more flexible than a material of the tubular member.

12. The medical elongated body according to claim 1, wherein a distal side of the shaft portion comprises a semicircular arc shape in an axially orthogonal cross section.

13. A medical elongated body comprising:

a shaft portion comprising a linear body and a proximal portion;

a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a resin material, and a second region having the inner layer and the outer layer, and a tubular member disposed between the inner layer and the outer layer in the second region, the proximal end

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opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and

a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer of the first region of the distal member.

14. The medical elongated body according to claim 13, wherein the tubular member has an opening portion in a side wall and wherein the outer layer of the second region is welded to the inner layer via the opening portion.

15. The medical elongated body according to claim 13, wherein an outer diameter of the first region is smaller than an outer diameter of the second region.

16. The medical elongated body according to claim 13, wherein the outer layer of the distal member has an inclined portion having an outer diameter decreasing from a distal end of the second region toward a distal side of the distal member.

17. The medical elongated body according to claim 16, wherein in an axially orthogonal cross section of the distal member, a wall thickness of the outer layer in the first region closer to the distal side than the inclined portion is the same as a wall thickness of the outer layer in the second region.

18. The medical elongated body according to claim 13, wherein a distal side of the shaft portion comprises a semicircular arc shape in an axially orthogonal cross section.

19. A medical elongated body comprising:

a shaft portion comprising a linear body and a proximal portion;

a distal member disposed in a distal side of the shaft portion and having a distal end and a proximal end, the distal member having a distal end opening portion adjacent to the distal end of the distal member and a proximal end opening portion adjacent to the proximal end of the distal member, the distal member including a lumen extending from the distal end opening portion to the proximal end opening portion, the distal member having a first region having an inner layer and an outer layer which are made of a first region resin material, and a second region having an inner layer and an outer layer which are made of a second region resin material, and a tubular member disposed between the inner layer and the outer layer in the second region, the tubular member has an opening portion, the outer layer of the second region is welded to the inner layer via the opening portion, the proximal end opening portion being disposed at a proximal end of the lumen, the proximal end opening portion comprising an inclined region having a proximally-decreasing dimension, and the proximal end opening portion of the distal member being located distal of the proximal portion of the shaft portion; and

a reinforcement layer comprising a metallic material arranged between the inner layer and the outer layer of the first region.

20. The medical elongated body according to claim 19, wherein the distal member has a third region, the third region has an outer diameter increasing from a distal side toward a proximal side of the inclined region.

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